

EP3120 - Statistical Mechanics

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Statistical mechanics of classical ideal gas - I

Classical ideal gas

- A classical gas is such that the De Broglie wavelength of particles is much less compared to the interparticle spacing

$$\lambda = \frac{h}{p} \ll \left(\frac{V}{N}\right)^{1/3} \quad (1)$$

- If the temperature of the the gas is T , we can estimate the typical thermal speed of the particles as

$$v_{th} = \sqrt{\frac{3k_B T}{m}} \quad (2)$$

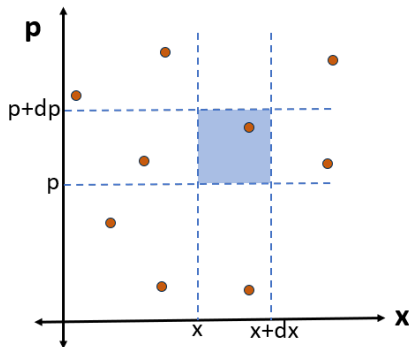
- Then the condition for classical gas becomes

$$\frac{h}{\sqrt{3k_B T m}} \left(\frac{N}{V}\right)^{1/3} \ll 1 \quad (3)$$

- Ideal gas means infinitesimal particles with no long range interaction and elastic collisions

Phase space

- We can determine all the physics of this gas by knowing the position and momentum of each particle at a time t
- This can be represented as points in a 6 dimensional phase space of position and momentum (for a 3D gas)
- For a 1D gas, the state of the gas can be represented by N points on a 2D phase space



Distribution function

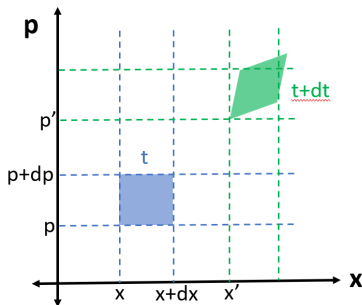
- We are not really interested in finding out the location of each particle in this phase space
- Rather we would like to build a probability density function (also called distribution function) $f(\mathbf{r}, \mathbf{p}, t)$
- This tells us the probability of finding **any** particle in an infinitesimal region of phase space around the point $(\mathbf{r}, \mathbf{p})$
- The number of particles in region $(\mathbf{r}, \mathbf{p}) \rightarrow (\mathbf{r} + d\mathbf{r}, \mathbf{p} + d\mathbf{p})$ at time t is $f(\mathbf{r}, \mathbf{p}, t)d\mathbf{r}d\mathbf{p}$
- More explicitly, in 1D the number of particles in the rectangular region $(x, p) \rightarrow (x + dx, p + dp)$ is $f(x, p, t)dxdp$
- When integrated over all of the phase space, we should get

$$\int d\mathbf{r} \int d\mathbf{p} f(\mathbf{r}, \mathbf{p}, t) = N \quad (4)$$

where N is the total number of particles

Phase space dynamics

- As the system evolves with time, the particle positions in phase space will change
- Under the action of an external force that depends only on the position, the infinitesimal phase space volume will evolve



- The particles in the blue region will move to the green region
- This means that

$$f(\mathbf{r} + \mathbf{v}\delta t, \mathbf{p} + \mathbf{F}\delta t, t + \delta t) d\mathbf{r}' d\mathbf{p}' = f(\mathbf{r}, \mathbf{p}, t) d\mathbf{r} d\mathbf{p} \quad (5)$$

Phase space dynamics

- The particles occupying the blue region will go to the green region
- H.W.: It can be shown that the area of the blue and green parallelograms are same for an external force that only depends on position
- This statement is generally true for other kinds of forces also (including internal forces which obey certain relations)
- This means that $d\mathbf{r}'d\mathbf{p}' = d\mathbf{r}d\mathbf{p}$ and so

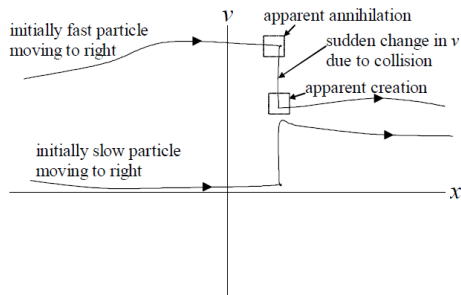
$$f(\mathbf{r} + \mathbf{v}\delta t, \mathbf{p} + \mathbf{F}\delta t, t + \delta t) = f(\mathbf{r}, \mathbf{p}, t) \quad (6)$$

- Doing a Taylor series expansion, dividing by δt , and taking limit $\delta t \rightarrow 0$ gives

$$\frac{\partial f}{\partial t} + \frac{\mathbf{p}}{m} \cdot \nabla_{\mathbf{r}} f + \mathbf{F} \cdot \nabla_{\mathbf{p}} f = 0 \quad (7)$$

Collisions

- However, if there are billiard-ball collisions, then this can change
- A collision would have the effect of suddenly making a particle disappear from one place in phase space and appearing at another place



- This would create some sudden change in f which is given as

$$\frac{\partial f}{\partial t} + \frac{\mathbf{p}}{m} \cdot \nabla_{\mathbf{r}} f + \mathbf{F} \cdot \nabla_{\mathbf{p}} f = \left(\frac{\partial f}{\partial t} \right)_{coll} \quad (8)$$

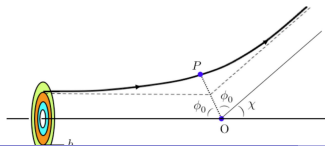
Modelling collisions

- Due to collisions, the positions don't change instantaneously, but the momentum does
- Consider a collision where two incoming particles with momenta $\mathbf{p}_1$ and $\mathbf{p}_2$ collide and go off with momenta $\mathbf{p}'_1$ and $\mathbf{p}'_2$
- $P_{12 \rightarrow 1'2'}$ is a probability density such that

$$\sigma = \int d\sigma = \int P_{12 \rightarrow 1'2'} d\mathbf{p}'_1 d\mathbf{p}'_2 \quad (9)$$

is the total cross section of incoming particles with momenta $\mathbf{p}_1, \mathbf{p}_2$ scattering into all possible values of $\mathbf{p}'_1 + d\mathbf{p}'_1, \mathbf{p}'_2 + d\mathbf{p}'_2$

- This $d\sigma$ is the area of the annular disk that leads to collision in this momentum range
- In classical mechanics it depends on the probability of a certain impact parameter (b) of collision between the two particles



Modelling collisions

- The probability of finding a target particle in the volume encountered by incident particle $\mathbf{p}_1$ in time δt becomes

$$f(\mathbf{r}, \mathbf{p}_2) d\mathbf{p}_2 u d\sigma \delta t \quad (10)$$

where $u = |\mathbf{v}_1 - \mathbf{v}_2|$ is the relative speed between colliding particles

- The probability of finding an incident particle is

$$f(\mathbf{r}, \mathbf{p}_1) d\mathbf{r} d\mathbf{p}_1 \quad (11)$$

- The probability of collision in time δt will be the product of the two, and that corresponds to the reduction in the probability of finding the particle at $\mathbf{p}_1$

$$\delta f_{12 \rightarrow 1'2'} d\mathbf{r} d\mathbf{p}_1 = -f(\mathbf{r}, \mathbf{p}_1) d\mathbf{r} d\mathbf{p}_1 \times f(\mathbf{r}, \mathbf{p}_2) d\mathbf{p}_2 P_{12 \rightarrow 1'2'} d\mathbf{p}'_1 d\mathbf{p}'_2 u \delta t \quad (12)$$

- The total change due to scattering into all possible final states becomes

$$\left. \frac{\partial f}{\partial t} \right|_{\text{out of } \mathbf{p}_1} = - \int d\mathbf{p}_2 d\mathbf{p}'_1 d\mathbf{p}'_2 f(\mathbf{r}, \mathbf{p}_1) f(\mathbf{r}, \mathbf{p}_2) u P_{12 \rightarrow 1'2'} \quad (13)$$

Scattering cross sections

- Calculation of the scattering probability will lead to an expression like

$$P_{12 \rightarrow 1'2'} = T_{12 \rightarrow 1'2'} \delta^4(\mathbf{p}_1 + \mathbf{p}_2 - \mathbf{p}'_1 + \mathbf{p}'_2) \quad (14)$$

- $T_{12 \rightarrow 1'2'}$ will come from the classical(quantum) scattering cross section and the Dirac-Delta function is there to ensure momentum and energy conservation, i.e.

$$\delta^4(\mathbf{p}_1 + \mathbf{p}_2 - \mathbf{p}'_1 + \mathbf{p}'_2) = \delta(p_{1x} + p_{2x} - p'_{1x} - p'_{2x}) \times \quad (15)$$

$$\delta(p_{1y} + p_{2y} - p'_{1y} - p'_{2y}) \times \quad (16)$$

$$\delta(p_{1z} + p_{2z} - p'_{1z} - p'_{2z}) \times \quad (17)$$

$$\delta(E_1 + E_2 - E'_1 - E'_2) \quad (18)$$

- Substituting this we get

$$\left. \frac{\partial f}{\partial t} \right|_{\text{out of } \mathbf{p}_1} = - \int d\mathbf{p}_2 d\mathbf{p}'_1 d\mathbf{p}'_2 f(\mathbf{r}, \mathbf{p}_1) f(\mathbf{r}, \mathbf{p}_2) u T_{12 \rightarrow 1'2'} \delta^4(\mathbf{p}_f - \mathbf{p}_i) \quad (19)$$

Modelling collisions

- Simplifying the notation a little bit, use $f_1 \equiv f(\mathbf{r}, \mathbf{p}_1, t)$

$$\left. \frac{\partial f_1}{\partial t} \right|_{\text{out of } \mathbf{p}_1} = - \int d\mathbf{p}_2 d\mathbf{p}'_1 d\mathbf{p}'_2 f_1 f_2 u T_{12 \rightarrow 1'2'} \delta^4(\mathbf{p}_f - \mathbf{p}_i) \quad (20)$$

- Similarly, the number of particles entering into the element $\mathbf{p}_1 \rightarrow \mathbf{p}_1 + d\mathbf{p}_1$ will be from all those collisions between $\mathbf{p}'_1$ and $\mathbf{p}'_2$ that end in particles with $\mathbf{p}_1$ and $\mathbf{p}_2$
- The rate of change of f due to this will be

$$\left. \frac{\partial f_1}{\partial t} \right|_{\text{into } \mathbf{p}_1} = \int d\mathbf{p}_2 d\mathbf{p}'_1 d\mathbf{p}'_2 f'_1 f'_2 u' T_{1'2' \rightarrow 12} \delta^4(\mathbf{p}_f - \mathbf{p}_i) \quad (21)$$

- where $u' = |\mathbf{v}'_1 - \mathbf{v}'_2|$, but for elastic identical particle collision, $u' = u$
- Since the microscopic process is time reversible, so

$T_{12 \rightarrow 1'2'} = T_{1'2' \rightarrow 12}$ and we have

$$\left. \frac{\partial f_1}{\partial t} \right|_{\text{coll}} = \int d\mathbf{p}_2 d\mathbf{p}'_1 d\mathbf{p}'_2 u (f'_1 f'_2 - f_1 f_2) T_{1'2' \rightarrow 12} \delta^4(\mathbf{p}_f - \mathbf{p}_i) \quad (22)$$

Boltzmann transport equation

- Putting together all the terms we get the Boltzmann transport equation

$$\frac{\partial f_1}{\partial t} + \frac{\mathbf{p}_1}{m} \cdot \nabla_{\mathbf{r}} f_1 + \mathbf{F} \cdot \nabla_{\mathbf{p}_1} f_1 = \int d\mathbf{p}_2 d\mathbf{p}'_1 d\mathbf{p}'_2 u(\mathbf{f}'_1 \mathbf{f}'_2 - f_1 f_2) \quad (23)$$

$$\times T_{1'2' \rightarrow 12} \delta^4(\mathbf{p}_f - \mathbf{p}_i) \quad (24)$$

- Assume that there is no external force and also that the distribution function is independent of position, i.e. $f = f(\mathbf{p}, t)$
- Under this condition the Boltzmann transport equation becomes

$$\frac{\partial f_1}{\partial t} = \int d\mathbf{p}_2 d\mathbf{p}'_1 d\mathbf{p}'_2 u(\mathbf{f}'_1 \mathbf{f}'_2 - f_1 f_2) T_{pp'} \delta^4(\mathbf{p}_f - \mathbf{p}_i) \quad (25)$$

- For simplicity using $T_{pp'} \equiv T_{1'2' \rightarrow 12}$

Boltzmann H theorem

- Define the Boltzmann function H

$$H = \int d\mathbf{p} f(\mathbf{p}, t) \log f(\mathbf{p}, t) \quad (26)$$

- It's time derivative is

$$\frac{dH}{dt} = \int d\mathbf{p} \frac{\partial f}{\partial t} [1 + \log f] \quad (27)$$

- The Boltzmann H theorem states that

$$\frac{dH}{dt} \leq 0 \quad (28)$$

- To prove this, substitute the Boltzmann transport equation for $\partial f / \partial t$

$$\frac{dH}{dt} = \int d\mathbf{p}_1 [1 + \log f_1] \int d\mathbf{p}_2 d\mathbf{p}'_1 d\mathbf{p}'_2 u(f'_1 f'_2 - f_1 f_2) T_{pp'} \delta^4(\mathbf{p}_f - \mathbf{p}_i) \quad (29)$$

$$= \int d\mathbf{p}_1 d\mathbf{p}_2 d\mathbf{p}'_1 d\mathbf{p}'_2 [1 + \log f_1] u(f'_1 f'_2 - f_1 f_2) T_{pp'} \delta^4(\mathbf{p}_f - \mathbf{p}_i) \quad (30)$$

Boltzmann H theorem

- Even if we interchange particle $\mathbf{p}_1$ with $\mathbf{p}_2$, still the relation should hold, giving

$$\frac{dH}{dt} = \int d\mathbf{p}_1 d\mathbf{p}_2 d\mathbf{p}'_1 d\mathbf{p}'_2 [1 + \log f_2] u(f'_1 f'_2 - f_1 f_2) T_{pp'} \delta^4(\mathbf{p}_f - \mathbf{p}_i) \quad (31)$$

- Adding these two expressions and dividing by 2, we get

$$\frac{dH}{dt} = \frac{1}{2} \int d\mathbf{p}_1 d\mathbf{p}_2 d\mathbf{p}'_1 d\mathbf{p}'_2 [2 + \log(f_1 f_2)] u(f'_1 f'_2 - f_1 f_2) T_{pp'} \delta^4(\mathbf{p}_f - \mathbf{p}_i) \quad (32)$$

- Now we could also reverse the collision and consider particles $\mathbf{p}'_1$ and $\mathbf{p}'_2$ colliding and going into $\mathbf{p}_1$ and $\mathbf{p}_2$
- By time-reversal symmetry of the collision $T_{pp'} = T_{p'p}$

Boltzmann H theorem

- So we get

$$\frac{dH}{dt} = \frac{1}{2} \int d\mathbf{p}_1 d\mathbf{p}_2 d\mathbf{p}'_1 d\mathbf{p}'_2 [2 + \log(f'_1 f'_2)] u(f_1 f_2 - f'_1 f'_2) T_{pp'} \delta^4(\mathbf{p}_f - \mathbf{p}_i) \quad (33)$$

- Adding these two expressions gives

$$\frac{dH}{dt} = \int d\mathbf{p}_1 d\mathbf{p}_2 d\mathbf{p}'_1 d\mathbf{p}'_2 [\log(f'_1 f'_2) - \log(f_1 f_2)] u(f_1 f_2 - f'_1 f'_2) T_{pp'} \delta^4(\mathbf{p}_f - \mathbf{p}_i) \quad (34)$$

$$= \int d\mathbf{p}_1 d\mathbf{p}_2 d\mathbf{p}'_1 d\mathbf{p}'_2 \left[\log \left(\frac{f'_1 f'_2}{f_1 f_2} \right) \right] (f_1 f_2 - f'_1 f'_2) u T_{pp'} \delta^4(\mathbf{p}_f - \mathbf{p}_i) \quad (35)$$

- The probability $T_{pp'} \geq 0$ and

$$\log \left(\frac{x}{y} \right) (y - x) \leq 0 \quad (36)$$

Boltzmann H theorem

- This means that the integrand is always less than or equal to 0 and that proves that

$$\frac{dH}{dt} \leq 0 \quad (37)$$

- The equality holds only if the integrand is always equal to zero, i.e.

$$f'_1 f'_2 = f_1 f_2 \quad (38)$$

$$\implies f(\mathbf{p}'_1, t) f(\mathbf{p}'_2, t) = f(\mathbf{p}_2, t) f(\mathbf{p}_1, t) \quad (39)$$

- This is under the condition that collisions go from $(\mathbf{p}_1, \mathbf{p}_2)$ to $(\mathbf{p}'_1, \mathbf{p}'_2)$
- In equilibrium conditions, we will have $dH/dt = 0$
- What will be the form of $f(\mathbf{p})$ in equilibrium?

Boltzmann distribution

- During collisions we have conserved quantities that are additive, i.e.

$$\mathbf{p}_1 + \mathbf{p}_2 = \mathbf{p}'_1 + \mathbf{p}'_2 \quad (40)$$

$$E_1 + E_2 = E'_1 + E'_2 \quad (41)$$

- If we have a distribution function $f(\mathbf{p})$ which is of the form of

$$f(\mathbf{p}) = \exp[g(\mathbf{p})] \quad (42)$$

- where $g(\mathbf{p})$ is some conserved quantity
 $g(\mathbf{p}_1) + g(\mathbf{p}_2) = g(\mathbf{p}'_1) + g(\mathbf{p}'_2)$, then that would satisfy the condition for equilibrium, i.e., $f(\mathbf{p}'_1, t)f(\mathbf{p}'_2, t) = f(\mathbf{p}_1, t)f(\mathbf{p}_2, t)$
- In fact, it can be shown that the only smooth function that has the property $f(x)f(y) = f(x+y)$ is $f(x) = e^{cx}$

Boltzmann distribution

- This equilibrium distribution is called as the Boltzmann distribution and it has the form

$$f_M = \exp \left[g(\mathbf{p}) \right] \quad (43)$$

- where

$$g(\mathbf{p}) = A + \mathbf{B} \cdot \mathbf{p} + Cp^2 \quad (44)$$

- The first term is a constant that is conserved by definition, the second term is momentum conservation, and the third term is energy conservation (non-relativistic)
- $A, \mathbf{B}, C$ are constant coefficients, and $\mathbf{B}$ is a vector for the momentum components. The Maxwell distribution becomes

$$f_M(\mathbf{p}) = A \exp \left[\mathbf{B} \cdot \mathbf{p} + Cp^2 \right] \quad (45)$$

Boltzmann distribution

- If the gas is stationary, it means that there is equal probability of a particle moving in any direction
- This means $B = 0$ and we have

$$f_M(\mathbf{p}) = A \exp(-Cp^2) \quad (46)$$

- Minus sign indicates a physically realistic distribution function
- To find A and C we use the normalization conditions that

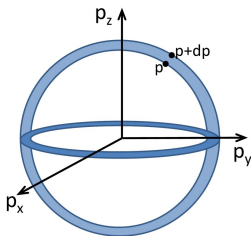
$$\int d\mathbf{p} f_M(\mathbf{p}) = N/V \quad (47)$$

$$\int d\mathbf{p} f_M(\mathbf{p}) \frac{p^2}{2m} = E/V \quad (48)$$

- Where E is the total energy of the gas

Normalization constant

- Assuming an isotropic distribution function, the phase space volume element can be considered as a spherical shell



$$\int_0^{\infty} 4\pi p^2 dp A e^{-Cp^2} = N/V \quad (49)$$

- Evaluate $I = \int_0^{\infty} dp p^2 e^{-p^2}$

Gamma integrals

- Substituting $t = p^2$ in I gives

$$I = \frac{1}{2} \int_0^\infty dt t^{1/2} e^{-t} = \frac{1}{4} \int_0^\infty t^{-1/2} e^{-t} dt \quad (50)$$

- Here we have used integration by parts. Substitute $t = x^2$

$$I = \frac{1}{2} \int_0^\infty e^{-x^2} dx = \frac{1}{4} \int_{-\infty}^\infty e^{-x^2} dx = \frac{1}{4} \int_{-\infty}^\infty e^{-y^2} dy \quad (51)$$

- Multiplying the two integrals (over x and y)

$$I^2 = \frac{1}{16} \int_{-\infty}^\infty dx \int_{-\infty}^\infty dy e^{-x^2-y^2} dx dy \quad (52)$$

- Converting from Cartesian to polar coordinates $x^2 + y^2 = r^2$,
 $dx dy = r dr d\theta$

$$I^2 = \frac{1}{16} \int_0^\infty r dr \int_0^{2\pi} d\theta e^{-r^2} \quad (53)$$

Gamma integrals

- Upon integrating we get

$$I^2 = \frac{\pi}{16} \implies I = \frac{\sqrt{\pi}}{4} \quad (54)$$

- Now we need to evaluate

$$4\pi A \int_0^\infty dp p^2 e^{-Cp^2} = 4\pi A C^{-3/2} \int_0^\infty dt t^2 e^{-t^2} = A\pi^{3/2} C^{-3/2} = N/V \quad (55)$$

- Generally such integrals can be evaluated by Gamma functions

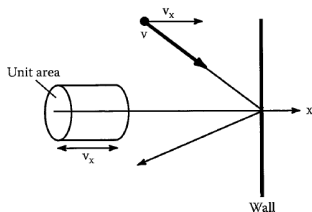
$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt \quad (56)$$

$$\Gamma(z+1) = z\Gamma(z) \quad (57)$$

$$\Gamma(n) = (n-1)! \quad (58)$$

$$\Gamma(1/2) = \sqrt{\pi} \quad (59)$$

Pressure



- The number of particles in the cylinder with velocity $\mathbf{v}$ is $f(\mathbf{p})v_x d\mathbf{p}$
- They collide with the wall in time v_x/v_x , i.e., in unit time, imparting a force $2mv_x f(\mathbf{p})v_x d\mathbf{p}$
- The pressure imparted by all the particles is then given by

$$P = \int_{v_x > 0} 2mv_x f(\mathbf{p})v_x d\mathbf{p} = m \int d\mathbf{p} v_x^2 f(\mathbf{p}) \quad (60)$$

$$= \frac{m}{3} \int d\mathbf{p} v^2 f(\mathbf{p}) \quad (61)$$

- Assuming isotropy in all directions

Pressure

- The pressure then becomes

$$P = \frac{1}{3m} \int d\mathbf{p} p^2 f(\mathbf{p}) = \frac{2}{3} \int d\mathbf{p} \frac{p^2}{2m} f(\mathbf{p}) = \frac{2E}{3V} \quad (62)$$

- From ideal gas law $P = Nk_B T/V$ which means $E = (3/2)Nk_B T$
- Now we can show that

$$\int d\mathbf{p} p^2 f(\mathbf{p}) = \int_0^\infty 4\pi p^2 dp p^2 A e^{-Cp^2} = 4\pi A \int_0^\infty dp p^4 e^{-Cp^2} \quad (63)$$

$$= 4\pi A \frac{1}{2C^{5/2}} \Gamma(5/2) = 4\pi A \frac{3\sqrt{\pi}}{8C^{5/2}} = \frac{3A}{2} \frac{\pi^{3/2}}{C^{5/2}} \quad (64)$$

- From this we get the condition

$$\frac{A}{C^{5/2}} = \frac{2Nk_B T m}{V\pi^{3/2}} \quad (65)$$

- And from the number density we had obtained

$$A C^{-3/2} = \frac{N\pi^{-3/2}}{V} \quad (66)$$

Maxwell Boltzmann distribution

- From these two relations, we get

$$C = \frac{1}{2k_B T m} \quad (67)$$

$$A = \frac{N}{V} \frac{1}{(2\pi m k_B T)^{3/2}} = \frac{n}{(2\pi m k_B T)^{3/2}} \quad (68)$$

- Here n is the number density
- The Maxwell-Boltzmann distribution becomes

$$f(\mathbf{p}) = \frac{n}{(2\pi m k_B T)^{3/2}} \exp \left[-\frac{p^2}{2m k_B T} \right] \quad (69)$$

- Exercise: Calculate the rms velocity of these gas particles

RMS velocity

- The root mean square velocity can be defined as

$$v_{rms}^2 = \frac{\int d\mathbf{p} df(\mathbf{p}) v^2}{N} \quad (70)$$

$$= \frac{V}{N} \int d\mathbf{p} f(\mathbf{p}) \frac{p^2}{m^2} = \frac{V}{Nm^2} \int d\mathbf{p} f(\mathbf{p}) p^2 \quad (71)$$

$$= \frac{V}{Nm^2} \frac{2Em}{V} \quad (72)$$

$$= \frac{3k_B T}{m} \quad (73)$$

- So the RMS velocity is $v_{rms} = \sqrt{\frac{3k_B T}{m}}$
- H.W: Calculate the average speed and the most probable speed